

Answer ALL questions.

Write your answers in the spaces provided.

- 1** Maize is grown as a source of starch.

The photograph shows a ripe maize cob.



(Source: © IGOR STEVANOVIC / SCIENCE PHOTO LIBRARY)

- (a) Describe the structure of starch.

(3)

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(b) In maize, starch is synthesised at an optimum temperature of between 20°C and 30°C.

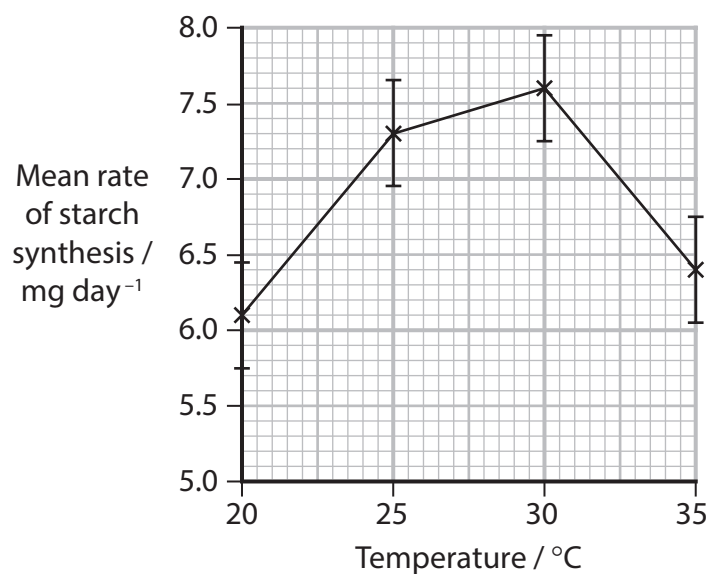
(i) Explain the effect this will have on the mass of starch produced in maize plants.

(2)

(ii) Studies have shown the effect of temperature on starch synthesis.

In an investigation, the rate of starch synthesis was measured at four different temperatures.

The results are shown in the graph.



Draw a table to show the results from the graph, excluding the standard deviations.

(3)

- (iii) Explain what the value for the standard deviation at 30°C means in this investigation.

(2)

Value of standard deviation

Explanation

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- (iv) Justify conclusions about the changes in mean rate of starch synthesis between 20°C and 25°C and between 25°C and 30°C.

(2)

20°C and 25°C

25°C and 30°C

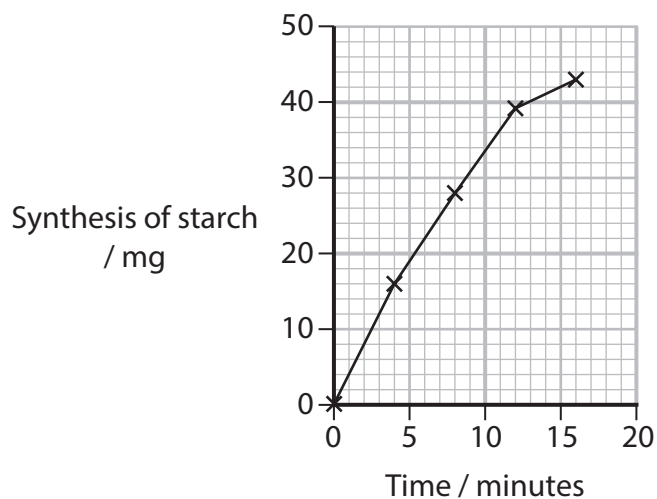
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(c) The graph shows the synthesis of starch at 30°C over a period of 16 minutes.



The table shows the initial rates of reaction calculated for four other temperatures.

Temperature / °C	Initial rate / mg min ⁻¹
20	2.0
25	2.6
30	
35	2.5
40	1.5

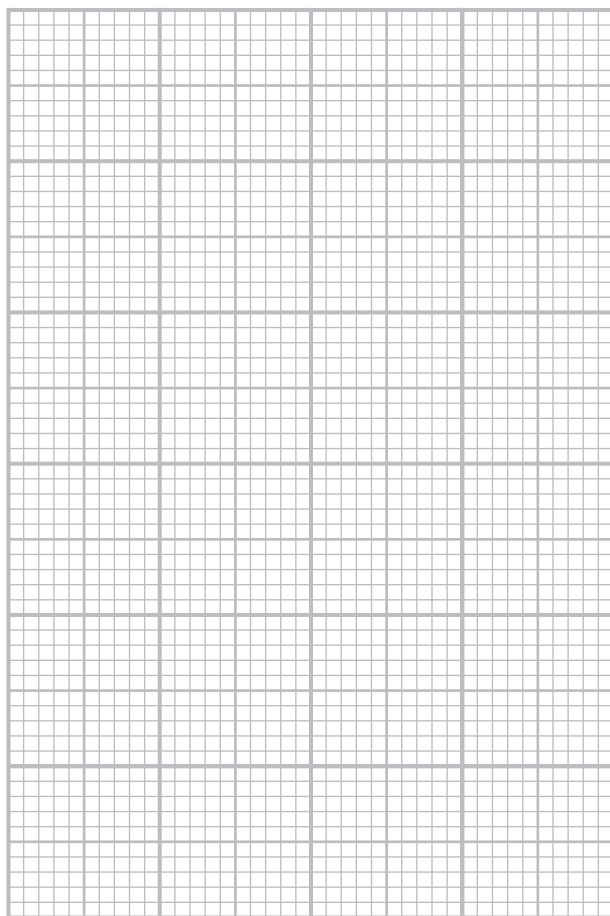
(i) Calculate the initial rate of reaction for 30°C shown in the graph.

(1)

Answer mg min⁻¹

(ii) Plot a graph of the data in the table and join the points with straight lines.

(4)



(iii) Give **two** reasons why the initial rate of reaction was measured.

(2)

1

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(Total for Question 1 = 19 marks)

